

Predicting MLB Draft Outcomes for NCAA Baseball Players Using XGBoost

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Introduction & Motivation

The MLB Draft (~600 picks, 20 rounds) determines which amateur players go professional. Draft likelihood is critical for college athletes deciding whether to enter the professional leagues or return to school.

This work contributes:

- 1. An XGBoost draft classifier (**88.2% accuracy**)
- 2. A college draft order regressor ($\rho = 0.700$)
- 3. A signing bonus predictor ($\rho = 0.734$)

Data Collection & Features

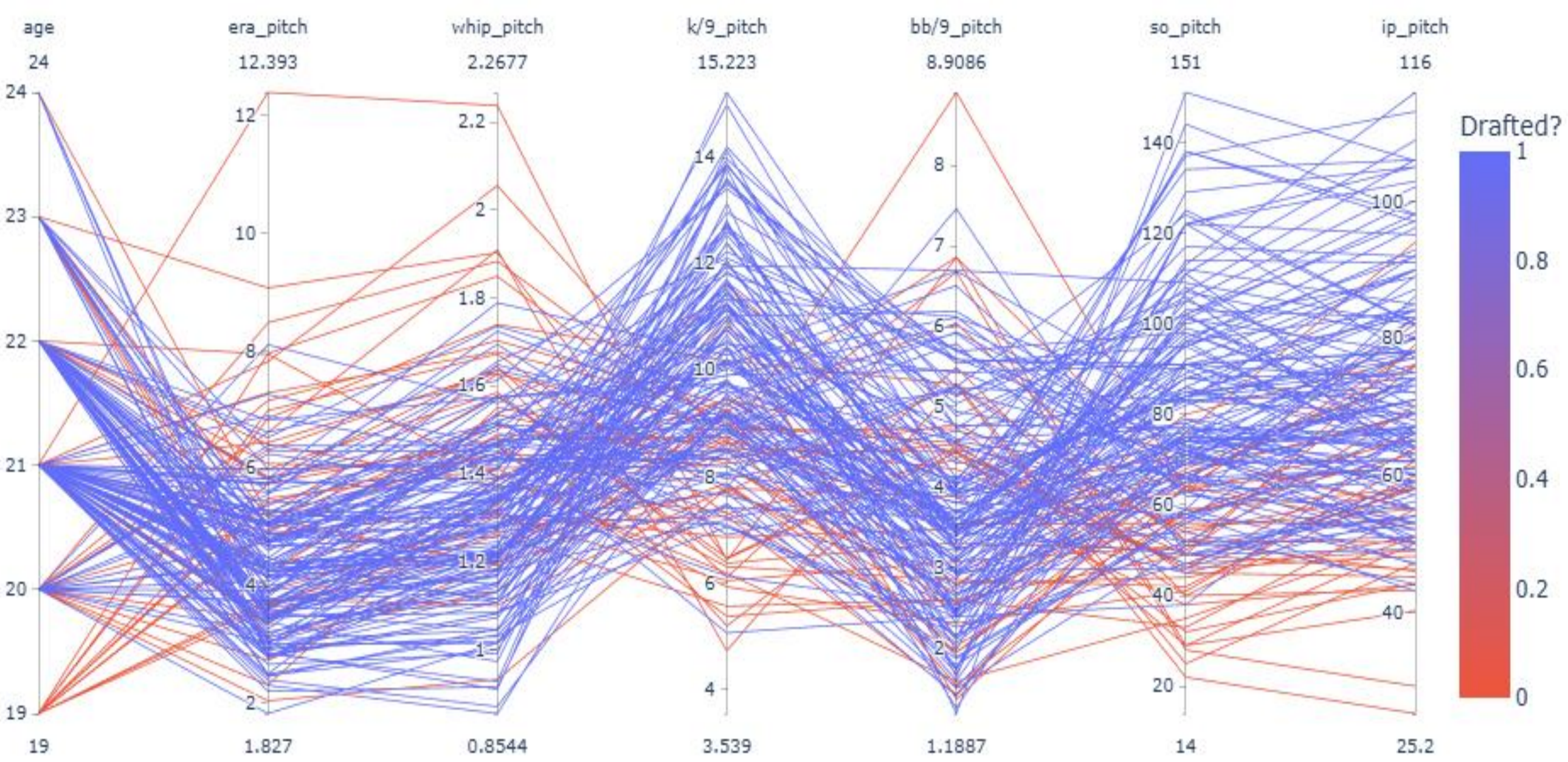
Data Source: ncaa_bbstats open-source Python package aggregating NCAA Stats, FanGraphs, and MLB Draft data.

Dataset: 16,728 player-seasons (1,438 drafted — 8.6%). Restricted to 2021–2025 to avoid equipment-era confounds and post-Covid changes.

81 Features:

- 33 player stats (ERA, WHIP, FIP, K%, wRC+, OPS)
- 48 team stats (win%, conf. RPI, Pythagorean expectation)
- Physical attributes (height, weight, handedness), only for drafted

Pitching Features by Predicted Draft Status



Methods

Three-Stage XGBoost Pipeline:

Stage 1 — Draft Probability Classifier

Binary classification on balanced data (undersampled to 1,438 per class). 80/20 stratified split.

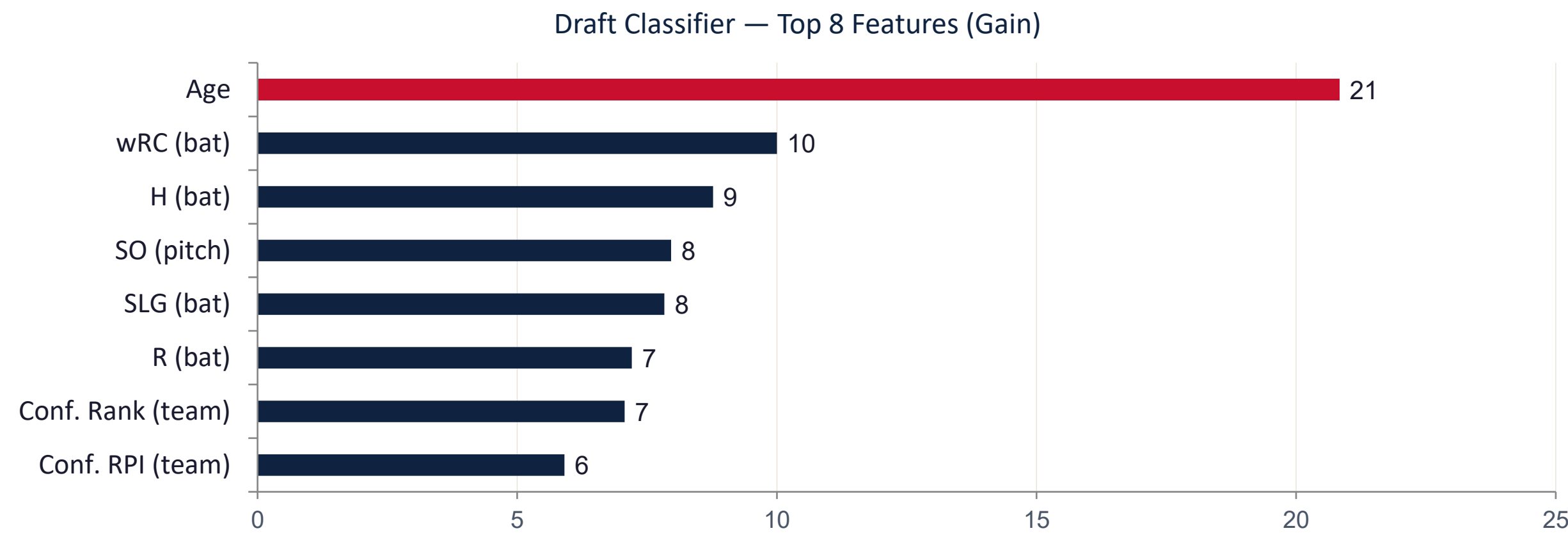
Stage 2 — College Draft Order Regressor

Ranks among college players only. Leave-one-year-out evaluation (train 2021–2024, test 2025).

Stage 3 — Signing Bonus Regressor

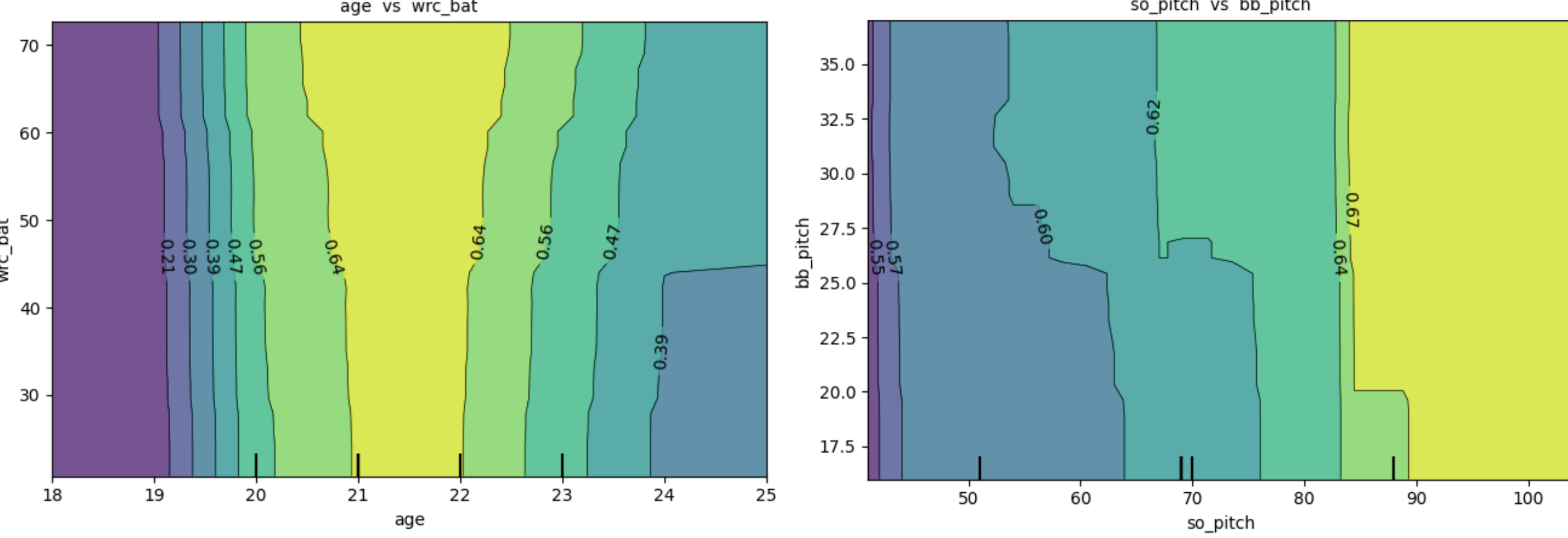
Log-transformed target with early stopping on validation MAE.

Feature Importance



Age dominates all three models. Batting metrics (wRC, H, SLG, R) and strikeout ability (SO) are strong secondary predictors. Team context (conf. RPI, conf. Rank) confirms scouts evaluate relative to competition.

Interaction PDP: Draft Probability



Future Work

1. Integrate with Prof. Gerber’s MLB draft → MiLB risks framework
2. Gather more data in upcoming years to compare trends
3. In a decade or so, look back and assess model’s accuracy in predicting if a player was under- or over-valued
4. Submit to the Journal of Sports Analytics

References

- [1] Gerber, Eric (2023). Competing Risks Analysis of MLB Draft Data.
- [2] Biggs, M (2025). ncaa_bbstats Python package (1. 1. 0). Zenodo.
- [3] Zhao, W., Akella, V. S., Yang, S., & Luo, X. (2025). Machine Learning in Baseball Analytics: Sabermetrics and Beyond. *Information*.
- [4] Penn, Matthew (2023). The Value of Playing in College Baseball to Major League Baseball Organizations. *Renée Crown University Honors Thesis Projects*.
- [5] Seneker, Derek (2025). Maximizing the MLB Draft: How College Performance and Leverage Impact Polayer Signing Bonus and Team Payoff. *CMC Senior Theses*.

Results

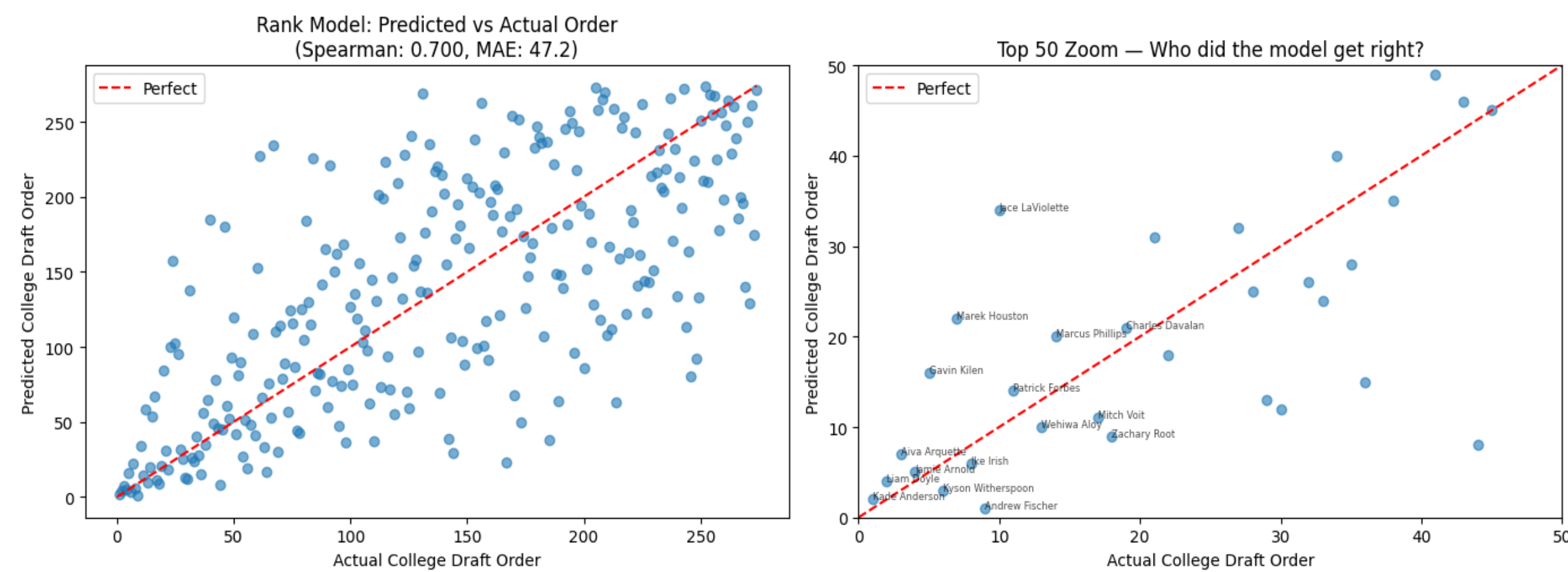
Draft Classifier (Balanced Dataset)

Class	Precision	Recall	F1	n
Not Drafted	0.901	0.858	0.879	288
Drafted	0.864	0.906	0.885	288

Draft Order Regressor (2025 Test)

Comparison	Spearman ρ	MAE
XGBoost Model	0.700	47.2
vs MLB Pipeline*	0.658 vs 0.841	17.3 vs 11.3

*98 matched players from MLB Pipeline Top 250



Signing Bonus Regressor (n = 273)

\$370K* Mean Absolute Error	0.448 R ² Score	0.734 Spearman Correlation
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*Signing bonus range: \$1,000 - \$9,250,000